HDOC Day-61

C Programming Activity - Two Sum Using Pointers

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **nums[]** containing positive integers, and an integer target. Pass the array to a function named **two_sum()** using its address and use pointers to access the array. Find two distinct indices **i** and **j** such that nums[i] + nums[j] = target. Display the indices in order. If no pair exists, display **-1 -1**.

C Program

```
#include <stdio.h>

void two_sum(int *nums, int n, int target)
{
    int i, j;

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (*(nums + i) + *(nums + j) == target)
            {
                printf("%d %d", i, j);
                return;
            }
        }
    }

    printf("-1 -1");
}

int main()
{
    int n, target;

    scanf("%d", &n);
    int nums[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    scanf("%d", &target);

    two_sum(nums, n, target);

    return 0;
}
```

Sample Input / Output

Input	Output

4 2 7 11 15 9	0 1

Explanation

The array name **nums** passes the address of its first element to **two_sum()**. The expressions ***(nums + i)** and ***(nums + j)** access array elements through pointer arithmetic. The nested loops test each distinct pair only once. When the required sum is found, the two indices are printed and the function returns. If no valid pair is found, the program prints **-1 -1**.